Impostor Phenomenon Moves Online: A Study of LinkedIn Usage and Impostor Phenomenon

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ABSTRACT

On LinkedIn, users portray an idealized image of themselves to others. In comparison, one could perceive themself as less skilled or deserving of their success, a characteristic of impostor phenomenon. This study, therefore, investigates the research question: "Is there a relationship between LinkedIn usage and impostor phenomenon?" and whether this relationship is mediated by social comparison. Participants (*N*=143) completed questionnaires—CIPS, INCOM, and a self-designed LinkedIn questionnaire—measuring the three variables. The results showed a small positive relationship between how successful participants perceived their connections to be and impostor phenomenon. A small negative relationship was also found between how long one has been on LinkedIn and impostor phenomenon which was fully mediated by social comparison. This could be due to those with a lower tendency for social comparison remaining on LinkedIn longer or more established users feeling less of a need to establish their self-concept via social comparison. The findings emphasize the importance of balanced conversations about the platform and more varied recruitment strategies. They also open up the field for further research.

1 INTRODUCTION

While in the past we could only compare our successes in the professional sphere to those of our colleagues or our social circle, with the widespread usage of business-oriented social media, particularly LinkedIn, we can now compare ourselves to anyone who creates an account. These ever-present options for comparison facilitated by social media have been shown to be associated with negative mental states, such as decreased self-esteem (Choukas-Bradley et al., 2018). Low self-esteem is also one of the characteristics of impostor phenomenon (Bravata et al., 2020).

Therefore, in this study, we will be looking at the relationships between LinkedIn usage, social comparison, and impostor phenomenon.

1.1 Impostor Phenomenon

First introduced by Clance & Imes in 1978, impostor phenomenon describes feelings of inadequacy and self-doubt about one's skills and achievements experienced despite overwhelming objective evidence to the contrary (Clance & Imes, 1978). Those who experience impostor phenomenon believe others overestimate their abilities and therefore are afraid of being exposed as frauds or as unworthy of their success (Bravata et al., 2020). When they compare themselves to others, they see them as more skilled or deserving of their accomplishments (Hutchins & Rainbolt, 2016). Rather than attributing their own achievements to internal factors they believe they come from external factors, such as luck, error, or good timing (Chandra et al., 2019). Even repeated success or positive feedback does not aid in weakening these feelings while negative feedback is used as proof of one's inadequacy (Clance & O'Toole, 1988).

To cope with feelings of impostor phenomenon individuals engage in excessive preparation, perfectionism, or procrastination (Jackson, 2018). Since they believe any mistake could expose their supposed incapability, they put off their work in fear of not being able to complete it to a high-enough standard (Chandra et al., 2019). Individuals with impostor phenomenon also avoid making risky decisions to not be faced with the possibility of failure (Jackson, 2018). These coping mechanisms along with the individuals' lower self-confidence compared to their peers result in negative life consequences. Among others, they lead to decreased well-being (Sonnak & Towell, 2001), prevent individuals affected by impostor phenomenon from reaching their full potential (Neureiter & Traut-Mattausch, 2016), and make them less satisfied with their job (Vergauwe et al., 2015).

1.2 LinkedIn Usage

LinkedIn is a business-oriented social media platform that facilitates matchmaking between job seekers and companies as well as discussion about work-related topics (Van Dijck, 2013). Launched in 2003, the platform currently has over 900 million users from over 200 countries (“About LinkedIn”) and 95% of recruiters use LinkedIn to find new talent (Ryan, 2020). Originally oriented mostly towards conversations between professionals, over time it shifted more towards a traditional social media interface with a newsfeed and updates on your connections (Van Dijck, 2013). Users’ profiles resemble CVs with sections for work experience, education, language and other skills, and recommendations (Zide et al, 2014).

Since the primary purpose of LinkedIn is to present oneself as an ideal candidate to recruiters and other professionals, impression management is applied when considering what to post on the platform (Roulin & Levashina, 2016). Therefore, similar to other social media, one's profile often becomes a reflection of an idealized self rather than a complex, real self (Van Dijck, 2013). What makes LinkedIn different is the salience of one’s profile being used as an assessment of their achievements and hence suitability for various positions, likely in comparison with other applicants, and the fact that the profile typically offers less information about oneself due to its structural nature (Van de Ven et al., 2017). These factors exacerbate the importance of presenting a well-polished image and can also facilitate easier comparison among users (Roulin & Levashina, 2016). Therefore, we believe that LinkedIn creates an environment that could interact with a number of characteristics of impostor phenomenon, and the relationship between the two is worth investigating.

1.3 Social Comparison

Social comparison theory was first outlined by Leon Festinger in 1974. It states that when an objective way to evaluate ourselves is not available we have a tendency to do so through comparison with others, especially those who are similar to us (Festinger, 1974). Such comparisons serve to reduce uncertainty and help us achieve more accurate self-knowledge (Wills & Sulls, 1991). What one learns through the process of social comparison can influence their self-esteem, subjective well-being, or aspirations (Suls et al., 2002). When one comes out negatively from the process, it can serve as a basis for self-improvement (Gibbons & Buunk, 1999). However, it can also cause one to conform with the group to achieve more favorable outcomes in future comparisons, or conversely to conceptualize those who come out better than them as undesirable 'others' (Festinger, 1974).

The relationship between social comparison and impostor phenomenon is twofold. On one hand, when one compares themself to others and finds themself lacking this can lead to feelings of impostor phenomenon (Fassl et al., 2020). On the other hand, people tend to engage in social comparison to various extents (Wheeler, 2000). Those who have low self-esteem, doubt their own capabilities, have an uncertain or unstable self-concept, or are under a lot of stress are particularly prone to do so (Campbell, 1990; Buunk & Ybema, 1995). Since those who experience impostor phenomenon generally have low self-esteem this can influence their propensity for social comparison (Chayerd & Bouffard, 2010). Due to this complex relationship between the concepts we wanted to investigate whether social comparison couldn't fully or partially explain the mechanism behind the potential relationship between LinkedIn usage and impostor phenomenon.

1.4 Comparison on Social Media

How the relatively novel phenomenon of social media interacts with our beliefs about ourselves, our self-esteem, and our need to compare ourselves to others has been a growing area of research (Vandenbosch et al., 2022; Faelens et al., 2021; Fardouly & Vartanian, 2016). Providing us with ubiquitous access to photos, opinions, and experiences of others they create an environment ripe with opportunities for comparison (Chou & Edge, 2012). The research in this area has largely focused on the effects of mainly picture-based social media on body image (Chou & Edge, 2012; Choukas-Bradley et al., 2018; Faelens et al., 2021). This literature has found that the more common occurrence of comparison combined with the selectively positive self-presentation of most social media users can lead to more comparison, more surveillance of oneself from others' perspectives, and lower self-esteem (Choukas-Bradley et al., 2018). However, other research has found that those who have a higher tendency to engage in social comparison use social media more heavily (Vogel et al., 2015).

Looking at the professional world, business-oriented social media, most popularly LinkedIn, have become widely used and are subject to the same bias towards positive self-presentation as other social media (Van Dijck, 2013). Given that social media can interact so negatively with our self-beliefs and self-esteem in the area of body image and beyond, when applied to the work context similar interactions might be occurring with feelings of impostor phenomenon which arise predominantly in the professional sphere. The relationship between various aspects of LinkedIn usage and impostor phenomenon is, therefore, fit for further analysis.

1.5 Present Study

Previous research has shown that LinkedIn users employ impression management strategies to appear more positively to others (Roulin & Levashina, 2016). Simultaneously, one of the key characteristics of impostor phenomenon is seeing others as more skilled and deserving of their success (Hutchins & Rainbolt, 2016). Therefore, being faced with this form of self-presentation on a work-oriented social media, which is the sphere where impostor phenomenon most commonly occurs, could interact with users' feelings of impostor phenomenon. This could occur through the mechanism of social comparison which in combination with social media often leads to negative consequences, such as decreased self-esteem or increased social media use (Faelens et al., 2021; Vogel et al., 2015).

This research will, hence, through a combination of questionnaires on LinkedIn usage, social comparison, and impostor phenomenon investigate whether there is a connection between these concepts. As impostor phenomenon is still a relatively underexplored phenomenon in the academic literature no research to date has looked at its connection to social media usage. However, with the world increasingly moving online, looking at how this interacts with and affects various psychological concepts, such as impostor phenomenon, is now more important than ever before. Hence, this novel study will address a gap between the rapidly developing world of social media and the impostor phenomenon literature.

The primary research question this study will investigate is the following: Is there a relationship between LinkedIn usage and impostor phenomenon? In light of the previously presented research, the experimental hypothesis is H_1: There is a relationship between LinkedIn usage and impostor phenomenon. while the null hypothesis is H_0: There is no relationship between LinkedIn usage and impostor phenomenon. There will also be a secondary research question that will look at whether social comparison can explain the mechanism behind the potential relationship between LinkedIn usage and impostor phenomenon. This secondary research question is the following: Is the relationship between LinkedIn usage and impostor phenomenon mediated by social comparison? Here, the experimental hypothesis is H_1: Social comparison mediates the relationship between LinkedIn usage and impostor phenomenon. while the null hypothesis is H_0: Social comparison does not mediate the relationship between LinkedIn usage and impostor phenomenon.

First, we will discuss the methodology of the study to see how we address these research questions including participants, measures, and procedure. Then, the results of primary and secondary analyses will be presented. These will be discussed in terms of their connections to the literature, implications, limitations, and directions for future research.

2 METHODOLOGY

2.1 Power Analysis

Prior to the commencement of the study, a power analysis was carried out to ensure the study would be sufficiently powered. Since no study has been conducted on this topic previously, the expected effect size was taken from a study on how social media presentation can make people believe others are happier and have better lives (Chou & Edge, 2012). Particularly, it was the effect size of a multiple linear regression of several measures of Facebook usage (e.g.: the number of hours spent on Facebook each week) and several demographic characteristics (e.g.: gender) on the perception that others are happier. This appeared to be the best approximation in the current literature of the type of feeling this study is trying to capture. The effect size (*f2*) of this multiple linear regression was 0.14. A power analysis determined that for a multiple linear regression model with 12 predictors (number of variables concerning LinkedIn being measured in the current study) at a 0.05 significance level to achieve 80% power to detect this effect, at least 122 participants are needed.

2.2 Participants

Participants (*N*=143) were LinkedIn users aged between 18 and 58 years (*M*=26.08, *SD*=8.43). From this 93 were students (65.03%) while 50 were non-students (34.96%). Participants' average length of education was 16.04 years (*SD*=2.25). More of the participants identified as women (*n*=91; 63.64%) than men (*n*=52; 36.36%). The majority of participants identified as White (*n*=94; 65.73%), 28 as Asian (19.58%), 6 as Black/Caribbean/African (4.20%), 9 as Mixed or multiple ethnic groups (6.29%), and 6 as Other (4.20%). Participants were recruited for a study supposedly on work habits through online dissemination of the questionnaire on social media, in private messages, and in various relevant interest groups. They were not compensated for their participation in any way. Informed consent was obtained from all participants and ethical approval was received from the London School of Economics (see Appendix A).

2.3 Measures

2.3.1 Impostor Phenomenon

Following previous studies on impostor phenomenon (Bravata et al., 2020), the Clance Impostor Phenomenon Scale (CIPS; Clance & O'Toole, 1988; see Appendix B) was used to measure feelings of impostor phenomenon. Permission to use this scale was obtained from its author. The CIPS is a scale validated through use in previous research that has very good internal reliability with reported Cronbach alpha values ranging from .85 to .96 (Mak et al., 2019). It provides participants with 20 statements related to feelings of impostor phenomenon and the participants are asked to indicate to what extent they agree with each statement using a 5-point Likert scale ranging from Not At All True (1) to Very True (5). Scores of ≤40 indicate few, scores between 41 and 60 indicate moderate, scores between 61 and 80 indicate frequent and scores over 80 indicate intense feelings of impostor phenomenon (Clance & O'Toole, 1988).

2.3.2 Social Comparison

The Iowa-Netherlands Comparison Orientation Measure (INCOM; Gibbons & Buunk, 1999; see Appendix C) was used to measure social comparison for the investigation of the secondary research question. This is also a scale validated through use in previous research with good internal reliability as evidenced by reported Cronbach alpha values ranging from .78 to .85 (Gibbons & Buunk, 1999). It opens with a short paragraph explaining that social comparison is neither good nor bad and that it occurs with different frequencies for different people. It then provides participants with 11 statements related to social comparison and the participants are asked to indicate to what extent they agree with each statement using a 5-point Likert scale ranging from I disagree strongly (A) to I agree strongly (E).

2.3.3 LinkedIn Usage

Demographic information (age, gender, ethnicity, occupation, education) and information on LinkedIn usage were collected using a self-designed questionnaire (see Appendix D). Since this is such a novel topic of research there are no prior questionnaires focused specifically on LinkedIn usage. While the created questionnaire was inspired by questions usually asked in questionnaires regarding general social media usage (e.g. Chou & Edge, 2012), these could not be used in their original form since LinkedIn, as a business-oriented social media, has some features that other social media sites do not and therefore these were not captured in the existing questionnaires. The self-designed questionnaire contains questions regarding how many years a participant has been using LinkedIn, how many hours per month they spend on LinkedIn, how many hours per month they spend doing specific activities on LinkedIn (browsing their newsfeed, finding new connections, looking at job postings, looking at other people's profiles, messaging with other people, posting, checking likes on their posts, updating information on their profile), whether they spend a significant amount of time doing any other activity on LinkedIn and if so what it is and how long they spend on it, how many connections they have on LinkedIn and whether they would say their connections are generally more, less or equally successful to them.

These three questionnaires were compiled into one in the following order: demographics and LinkedIn usage questionnaire followed by the INCOM and finally the CIPS.

2.4 Procedure

The study was conducted online through the cloud-based research platform, Gorilla. Participants received the study link and upon opening it were first shown a page containing information about the study background, procedure, risks and benefits of taking part, their rights as a participant, and the organizers behind the research. Slight deception was employed on this page since participants could not be fully informed of the aim of the study prior to filling in the questionnaires as this could affect their responses. Therefore, they were initially told the study was investigating work habits. However, as much information as possible (without influencing their responses) was revealed to them beforehand. The true aim was then revealed, and everything was explained on the debrief page after the completion of the questionnaires. On the initial information page, they also generated their anonymous participant code which they could use to contact the researchers at a later point if needed.

After this page, the participants were asked to provide their consent to taking part in the study by checking six mandatory checkboxes. They could also choose to exit the study at this point which would result in their exclusion from the study. The next page contained the demographic information questionnaire (collecting data on age, gender, ethnicity, occupation, and education) and the question whether they are a LinkedIn user. The participants who chose 'No' from the dropdown in response to this last question or did not answer it were excluded from the study. Those who chose 'Yes' then saw a page with further questions regarding LinkedIn usage (as outlined above). This was followed by another page containing the Iowa-Netherlands Comparison Orientation Measure. On the next, they saw the Clance Impostor Phenomenon Scale. Finally, participants saw a debrief screen with information about the true aim of the study, relevant resources if the study caused them any distress, and points of contact if they wanted to withdraw their data or had any questions.

3 RESULTS

3.1 Preliminary Analyses

Initially, the data was cleaned from all unnecessary columns leaving only the Participant IDs and the participants' responses to each question on the three questionnaires. It was then converted from a long to a wide format so data for every participant was in one row. Initially, data for 184 participants were available. Then, a listwise deletion strategy was employed which excluded any participants who didn't answer more than four questions (>10%; n=41) from the analysis. For the remaining missing data mean imputation was used. An impostor score was calculated for each participant as a sum of all their responses to the CIPS questions. Similarly, a social comparison score was also calculated for each participant by adding all their responses to the INCOM questions together with questions 5 and 11 being reverse-coded. The higher these scores are the more that participant experiences impostor phenomenon or social comparison respectively.

Descriptive statistics for all the quantitative variables regarding LinkedIn usage and the impostor score as well as their correlations can be found in Appendix E.

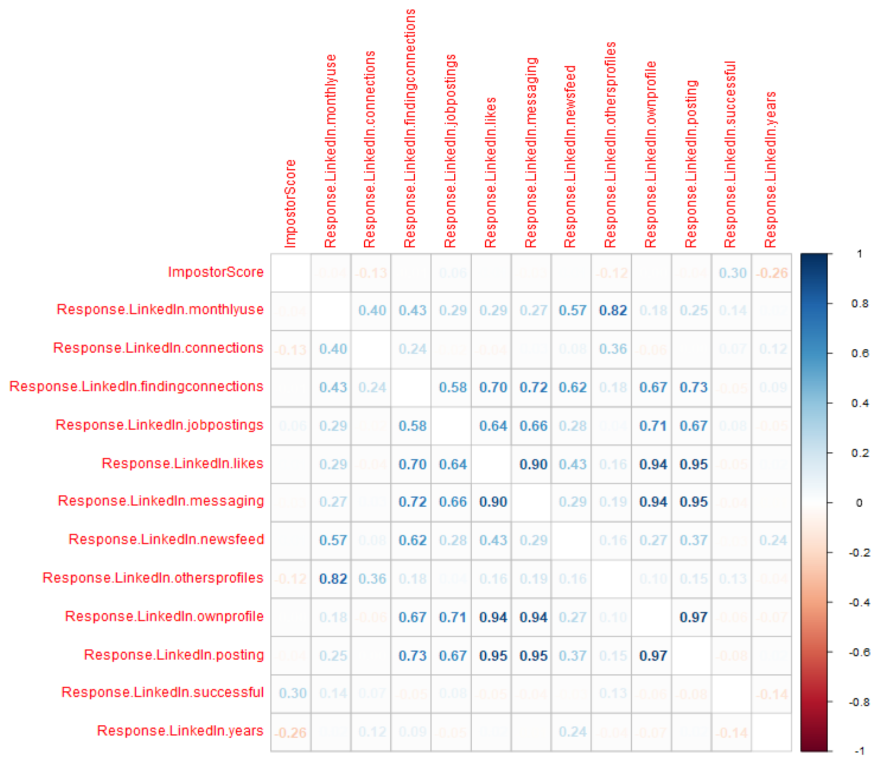

Figure 1: Correlation Matrix of All Possible Predictors

The correlations are also shown in the color-coded correlation plot above where higher correlations are shown as darker. From this analysis we can see significant correlations between the variables measuring time spent on particular activities on LinkedIn (posting, browsing newsfeed, checking likes, etc.) per month and the overall measure of time spent on LinkedIn monthly. To avoid issues of multicollinearity and overfitting in line with the strategy set out in the study's pre-registration highly-correlated variables were removed from the primary analysis.

3.2 Primary Analysis

The primary analysis consists of a multiple linear regression with participants' answers to questions regarding LinkedIn usage serving as predictor variables and the participants' impostor score as the outcome variable. This analysis was chosen as we had multiple quantitative, normally distributed measures of LinkedIn usage and we wanted to investigate their relationship with another quantitative variable, namely impostor score (Leech et al., 2003). Multiple regression is useful in that it lets us establish the unique effect of each of the predictors on the outcome variable since it controls for the influence of the other predictors by eliminating their common variance (Urdan, 2010). Additionally, our data sufficiently met all prerequisites for carrying out multiple regression, including linearity, normality of residuals, lack of collinearity, and homogeneity of variance. This can be seen in the graphs in Appendix F.

Following the correlation analysis, only the overall measure of time spent on LinkedIn was included in the multiple regression rather than measures for individual activities. This results in four predictors: monthly use in hours, number of connections, number of years they have been using LinkedIn, and whether they think their LinkedIn connections are more, less, or equally as successful as them. The descriptive statistics of these variables as well as the multiple regression output can be found in the tables below.

Variable	M	SD	1	2	3	4
1. ImpostorScore	64.20	15.91				
2. Response.LinkedIn .monthlyuse	8.10	14.64	-.04 [-.20, .13]			
3. Response.LinkedIn .connections	448.94	705.82	-.13 [-.29, .04]	.40** [.26, .53]		
4. Response.LinkedIn .successful	0.54	0.51	.30** [.14, .44]	.14 [-.03, .30]	.07 [-.10, .23]	
5. Response.LinkedIn .years	4.59	4.03	-.26** [-.41, -.10]	.02 [-.14, .19]	.12 [-.04, .28]	-.14 [-.30, .02]

Note. M and SD are used to represent mean and standard deviation, respectively. Values in square brackets indicate the 95% confidence interval for each correlation. The confidence interval is a plausible range of population correlations that could have caused the sample correlation (Cumming, 2014). * indicates $p < .05$. ** indicates $p < .01$.

Table 1: Descriptive Statistics and Correlations of Predictor and Output Variables

Predictor	b	b 95% CI [LL, UL]	$beta$	$beta$ 95% CI [LL, UL]	sr^2	sr^2 95% CI [LL, UL]	r	Fit
(Intercept)	64.55**	[59.64, 69.47]						
Response.LinkedIn.monthlyuse	-0.03	[-0.22, 0.15]	-0.03	[-0.20, 0.14]	.00	[-.01, .01]	-.04	
Response.LinkedIn.connections	-0.00	[-0.01, 0.00]	-0.11	[-0.28, 0.06]	.01	[-.02, .04]	-.13	
Response.LinkedIn.successful	8.76**	[3.87, 13.66]	0.28	[0.12, 0.44]	.08	[-.00, .16]	.30**	

Response.LinkedIn.years	-0.81*	[-1.43, -0.19]	-0.21	[-0.36, -0.05]	.04	[-.02, .10]	-.26**	
								R^2 = .153** 95% CI[.04,.24]

Note. A significant *b*-weight indicates the beta-weight and semi-partial correlation are also significant. *b* represents unstandardized regression weights. *beta* indicates the standardized regression weights. sr^2 represents the semi-partial correlation squared. *r* represents the zero-order correlation. *LL* and *UL* indicate the lower and upper limits of a confidence interval, respectively.
* indicates $p < .05$. ** indicates $p < .01$.

Table 2: Multiple Regression Output

There was a significant relationship ($p<0.01$) between whether one thought their LinkedIn connections are more, less, or equally as successful as them ($M = 0.54$, $SD = 0.51$) and their score on the CIPS ($M = 64.20$, $SD = 15.91$). For each one point increase in success perception, meaning from less successful to equally successful (from -1 to 0) or from equally successful to more successful (from 0 to 1), there was a 8.76 point increase (95% CI: 3.87, 13.66) in the participant's impostor score controlling for the other variables. The standardized effect was 0.28 (95% CI: 0.12, 0.44). There was also a significant relationship ($p<0.05$) between how many years one has been using LinkedIn ($M = 4.59$, $SD = 4.03$) and their score on the CIPS. For each one year increase in length of LinkedIn usage, there was a 0.81 point decrease (95% CI: -1.43, -0.19) in the participant's impostor score controlling for the other variables. The standardized effect was -0.21 (95% CI: -0.36, -0.05). No statistically significant relationship was found between how many hours per month one spends on LinkedIn and their CIPS score or between how many connections one has on LinkedIn and their CIPS score. The overall *R2* value of the model was .153 (95% CI: .04, .24) significant at the $p<0.01$ level. This means 15.3% of the variation in participants' impostor scores is explained by this model. The F-ratio is 6.242 with an associated p-value of $p<0.01$ meaning the model fits the data better than an empty model, a model without any predictors that simply predicts the mean of the outcome variable every time.

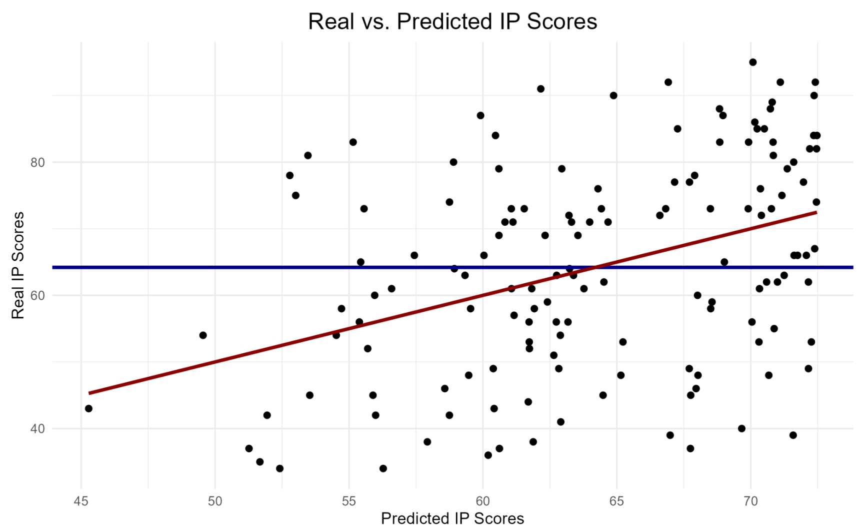

Figure 2: Plot of Multiple Linear Regression

This notion is further supported by the above plot where we can see the predictions of an empty model indicated by the blue line and the predictions made by our model with four predictors indicated by the red line. Since the red line captures the variance of the data points more accurately we can say that our model fits the data better than an empty model. Therefore, we have found evidence that supports our experimental hypothesis for our primary research question since there is a relationship between some aspects of LinkedIn usage and impostor phenomenon.

3.2 Secondary Analyses

The analysis of the secondary research question consisted of a mediation model where the number of years a person has been using LinkedIn (representing LinkedIn usage) was the predictor, the participant's score on the Clance Impostor Phenomenon Scale was the outcome variable and the participant's score on the Iowa-Netherlands Comparison Orientation Measure was the mediator. This analysis was chosen based on the interaction between social comparison and impostor phenomenon which has been shown in the literature and a theorized interaction between social comparison and LinkedIn usage. Mediation allows us to explore whether these interactions with social comparison could explain the mechanism behind the relationship between LinkedIn usage and impostor phenomenon (MacKinnon et al., 2007).

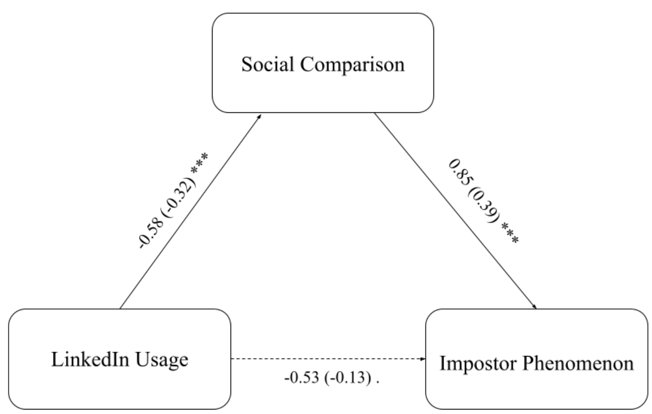

*Figure 2 - Social Comparison Moderation Model (*** indicates p<0.001, . indicates p<0.1)*

The *a* path measuring the effect of LinkedIn usage on social comparison had a size of -0.58 with an associated *p*-value of $p<0.001$. This means that for every additional year a participant used LinkedIn ($M = 4.59$, $SD = 4.03$) there was a 0.58 point decrease in their score on the INCOM ($M = 38.41$, $SD = 7.31$). The standardized effect size was -0.32. The *b* path measuring the effect of social comparison on impostor phenomenon controlling for LinkedIn usage had the size of 0.85 with an associated *p*-value of $p<0.001$. Therefore, for every additional point on the INCOM there was an 0.85 point increase in the participant's score on the CIPS ($M = 64.20$, $SD = 15.91$). The standardized effect size for this path was 0.39. The *c'* path measuring the effect of LinkedIn usage on impostor phenomenon controlling for social comparison was -0.53 with an associated *p*-value of $p<0.1$. Meaning that for every additional year a participant has been using LinkedIn there is a 0.53 point decrease in their CIPS score. The standardized effect size for the *c'* path was -0.13. As we can see the effect of LinkedIn usage on impostor phenomenon decreased in this model compared to the multiple regression model both in size and in statistical significance. Since the direct *c'* path is no longer significant at the $p<0.05$ level we can say that this is a full mediation. Simultaneously, as the paths from LinkedIn usage to social comparison and from social comparison to impostor phenomenon are statistically significant we can say that we found evidence to support the experimental hypothesis for our secondary research question.

Exploratory analysis was also carried out to see whether any of the demographic characteristics collected about the participants (age, gender, ethnicity, occupation, education) act as moderators for the relationship between LinkedIn usage and impostor phenomenon. However, none of the moderation effects were found to be statistically significant. The effect for age was -0.03 with an associated *p*-value of 0.39, for gender the effect was 0.32 with an associated *p*-value of 0.62, for ethnicity the effect was 0.55 with an associated *p*-value of 0.24, for occupation the effect was -0.23 with an associated *p*-value of 0.79, and for education the effect was 0.23 with an associated *p*-value of 0.17. The tables with more detailed outputs of each moderation model can be found in Appendix G.

4 DISCUSSION

4.1 Summary of Present Study

Previous research has found that one of the core characteristics of impostor phenomenon is feeling like others are more skilled or deserving of their success (Hutchins & Rainbolt, 2016). At the same time, social media offers increased opportunities for comparison with others' profiles carefully curated to highlight the positive aspects of their lives (Roulin & Levashina, 2016). This has been shown to interact negatively with people's self-confidence in other areas, such as body image (Choukas-Bradley et al., 2018). Since impostor phenomenon predominantly occurs in the work sphere, we believe it could interact with the usage of business-oriented social media, the most popular of which is LinkedIn. Therefore, the present study set out to investigate the relationships between LinkedIn usage, social comparison, and impostor phenomenon. To do so we collected data through the Clance Impostor Phenomenon Scale, the Iowa-Netherland Comparison Orientation Measure, and a self-designed LinkedIn questionnaire measuring various aspects of users' utilization of the platform.

The results showed that the length of LinkedIn usage has a small negative effect on impostor phenomenon while the perception of their connections' success has a small positive effect. Therefore, we found evidence to support that there is a relationship between LinkedIn usage and impostor phenomenon. Results of the mediation of the relationship between LinkedIn usage and impostor phenomenon by social comparison showed that the mediation was significant with a medium negative effect between LinkedIn use and social comparison and a medium positive effect between social comparison and impostor syndrome. The direct effect between LinkedIn usage and impostor phenomenon was no longer significant at the $p<0.05$ level. Here we, therefore, found evidence to support that social comparison mediates the relationship between LinkedIn usage and impostor phenomenon.

4.2 Connection to Literature

The mediation analysis shows us that those who have been using LinkedIn for longer have a lower tendency to engage in social comparison, and thus fewer feelings of impostor phenomenon. While the relationship between the latter two has been recorded in the literature previously (Fassl et al., 2020), the one between the former two is novel. One explanation behind this could be that those who engage in less social comparison are not as negatively affected by the constant opportunities for comparison and the impression management techniques of other users on LinkedIn and therefore manage to remain on the platform for longer. Alternatively, research states that in uncertain times when one is unsure about their self-concept they are more likely to engage in social comparison (Campbell, 1990). Individuals who have been using LinkedIn for longer are more likely to be and feel more established either in their careers more generally or within the LinkedIn environment specifically. Therefore, they experience less of a need to establish knowledge about themselves through means of social comparison.

This finding is, however, in contrast with findings from similar studies, for example concerning the relationship between Facebook usage and perceptions of other people's lives. Here the researchers found that the number of years one used Facebook was positively correlated with the beliefs that others have a better life and that others are happier, and negatively correlated with the belief that life is fair (Chou & Edge, 2012). Hence, those who had been using the social media platform for longer experienced worse mental states. Differences in the scope of the two social media platforms and the examined beliefs could provide an explanation. On Facebook, users can access information on various areas of others' lives from work and school through love life and friends to travel and hobbies. Therefore, it might be easier for them to find and focus on the things that others have better than them or areas in which they are happier. Meanwhile, LinkedIn is only focused on the professional area meaning users who have a lower tendency for social comparison or are more secure in this area have fewer instances when they can compare themselves and thus experience fewer feelings of impostor phenomenon.

The other significant effect was that of success perception. Given that seeing others as more successful than you is one of the characteristics of impostor phenomenon this is in line with previous research on the topic (Hutchins & Rainbolt, 2016; Hawley, 2019) and lends itself to two explanations. Either those with higher levels of impostor phenomenon reported their connections as more successful due to the warped perception created by the phenomenon or when people surround themselves on LinkedIn with those they perceive as more successful they are more likely to feel like they don't deserve to be where they are and hence experience impostor phenomenon.

4.3 Implications

Since this research is correlational the implications of its findings are twofold. On one hand, LinkedIn users should be more aware of the potential impacts their usage of the platform can have on their feelings of impostor phenomenon. LinkedIn is promoted as a beneficial or even necessary tool for finding a job whether that be at employment workshops, on business-oriented websites, or simply among colleagues (Sharone, 2017; Griffin, 2021). However, the negative aspects of its use are rarely mentioned. Therefore, awareness should be raised about the complex feelings and mental states that can arise from the usage of the platform. This can be achieved through social media literacy programs (Jeong et al., 2012; Cho et al., 2022), more balanced articles offering multiple perspectives on the matter, or more honest conversations about the platform in general. That way users could understand that they are not alone in the experience but rather that it is more likely to occur with certain aspects of the platform's usage, such as having connections one perceives as more successful or using the platform for a small number of years. This can help them find ways to mitigate or at least account for these feelings.

On the other hand, for users who experience more feelings of impostor phenomenon this might impact their experience on the platform, remaining on it for a shorter amount of time or predominantly perceiving their connections as more successful than them. This means their chances of recruitment might be impacted as they might not be on the platform long enough for recruiters to notice them or employers might take their later inactivity on the platform as a negative sign. In line with its vision of accessibility ("Practicing Responsibly") LinkedIn itself might want to address this issue. It could do so by investigating how its structures could be adjusted to offer an equally enjoyable experience for all its users including this group of users. If the platform does not decide to do so on its own perhaps this could be achieved by lobbying from its users or even governmental requirements. Additionally, employers might want to proactively address this potential bias in recruitment through LinkedIn by utilizing a wider range of recruitment strategies beyond the platform.

4.4 Limitations and Future Directions

The limitations of this study include the fact that since none of the variables were experimentally manipulated the design of the study is correlational and thus no causal inferences can be drawn from this data.

Furthermore, this study deals with internal states and processes, such as impostor phenomenon and social comparison, which are difficult to access through other means than self-report. However, self-report tends not to be reliable because participants may not be able to or want to represent their internal states accurately. This type of data collection can suffer from social desirability bias where participants answer in ways that are more socially desirable than their true answer (Grimm, 2010), such as indicating that they experience less impostor phenomenon or social comparison than they really do. Self-report data also suffers from demand effects where the participants infer (correctly or incorrectly) what the true aim of the study is and adjust their responses accordingly (Grimm, 2010). However, where possible established scales, such as those used to access internal concepts in this study, try to limit the occurrence of these biases with the wording of their questions, and slight deception was employed to conceal the true aim of the study.

Additionally, there have been some issues raised regarding the ability of the CIPS to fully and accurately capture the concept of impostor phenomenon (French et al., 2008). It has, however, been regarded as better than other scales used for this purpose (Holmes et al., 2010; Chrisman et al., 1995) and therefore was selected as the instrument of choice for this study.

The use of a self-designed questionnaire to measure LinkedIn usage means there is no evidence for the reliability of this measure. However, since it measures behavior rather than internal processes and since its questions are mostly based on other previously established questionnaires measuring social media usage, we believe it should be sufficient in accurately measuring the desired concept.

There are also limitations related to the sample as this was obtained through opportunity sampling and therefore is not representative of the general LinkedIn user base. Firstly, with a mean age of 26.08, our sample is younger than typical LinkedIn users since 60% of the platform's user base is in the 25-34 age range with only 21.7% in the 18-24 age range (Dixon, 2023a). Given the mean age, students are likely overrepresented in this study's sample with 65.03% and an average length of education of 16.04 years. This indicates that the sample was composed largely of master's and PhD students (since this study was conducted in the UK where primary+secondary education is 12 years, a bachelor's degree is 3 years and a master's degree is 1 year). However, only 23% of LinkedIn users in the US hold a master's degree or equivalent (Dixon, 2023c). Additionally, our sample is overly female with 63.64% women while the general LinkedIn population is predominantly male with only 43.7% of women (Dixon, 2023b). Data on the ethnicity statistics of LinkedIn users are harder to find but the predominantly White (65.73%) constitution of our sample seems to be more-or-less accurate (Campbell, 2023). These differences between this study's sample and the general LinkedIn population could influence the study's ability to capture trends among LinkedIn users accurately. Furthermore, it decreases its generalizability to the wider LinkedIn population.

In terms of future directions, further studies should be conducted that would test the relationships between LinkedIn usage, social comparison, and impostor phenomenon experimentally so that causation could be inferred. Additionally, further studies should try to recruit representative samples to increase the generalizability of the results. This study serves as the foundation for research into a previously unexplored field. Therefore, much work remains to be done to capture the nuances of the relationships between these variables. As technology continues to develop at a rapid pace research needs to embrace the novel avenues of exploration this offers up in order to be able to keep up.

5 CONCLUSION

This study tackled the primary research question: “Is there a relationship between LinkedIn usage and impostor phenomenon?” and the secondary research question: “Is this relationship mediated by social comparison?”. It found that users who used LinkedIn for a higher number of years experienced fewer feelings of impostor phenomenon which was fully mediated by them having a lower tendency to engage in social comparison. Additionally, it found that as users’ perception of the success of their connections in relation to them increased so did their feelings of impostor phenomenon. This could mean that these aspects of LinkedIn usage affect the extent to which users experience impostor phenomenon or that those who experience impostor phenomenon to different extents experience the platform differently.

To tackle this, we suggest more honest discussion of the complex feelings that accompany LinkedIn usage should be included at workshops, in articles, or in conversations where LinkedIn is being recommended. Additionally, LinkedIn and the recruiters who utilize it should consider the potential bias this could be introducing into their recruitment process. As social media becomes increasingly prolific in our everyday lives these findings emphasize the importance of considering the ways its usage can negatively interact with users’ mental states beyond body image which we most commonly see studied. At the same time, they serve as the foundation for further research into this underexplored area.

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Appendix A

Ethics Approval from the London School of Economics

Research Ethics

Dear Rebeka PRIVOZNIKOVA,

The ethics review application for the project:

Impostor Phenomenon Moves Online: a study of Linke
Ref: 147027

has been **approved**.

Appendix B

Clance Impostor Phenomenon Scale

Please indicate how much you agree with each statement below, by using the scale provided.

I have often succeeded on a test or task even though I was afraid that I would not do well before I undertook the task.

Not At All True	1	2	3	4	5	**Very True**

I can give the impression that I'm more competent than I really am.

Not At All True	1	2	3	4	5	**Very True**

I avoid evaluations if possible and have a dread of others evaluating me.

Not At All True	1	2	3	4	5	**Very True**

When people praise me for something I've accomplished, I'm afraid I won't be able to live up to their expectations of me in the future.

Not At All True	1	2	3	4	5	**Very True**

I sometimes think I obtained my present position or gained my present success because I happened to be in the right place at the right time or knew the right people.

Not At All True	1	2	3	4	5	**Very True**

I'm afraid people important to me may find out that I'm not as capable as they think I am.

Not At All True	1	2	3	4	5	**Very True**

I tend to remember the incidents in which I have not done my best more than those times I have done my best.

Not At All True	1	2	3	4	5	**Very True**

I rarely do a project or task as well as I'd like to do it.

Not At All True	1	2	3	4	5	**Very True**

Sometimes I feel or believe that my success in my life or in my job has been the result of some kind of error.

Not At All True	1	2	3	4	5	**Very True**

It's hard for me to accept compliments or praise about my intelligence or accomplishments.

Not At All True	1	2	3	4	5	Very True

At times, I feel my success has been due to some kind of luck.

Not At All True	1	2	3	4	5	Very True

I'm disappointed at times in my present accomplishments and think I should have accomplished much more.

Not At All True	1	2	3	4	5	Very True

Sometimes I'm afraid others will discover how much knowledge or ability I really lack.

Not At All True	1	2	3	4	5	Very True

I'm often afraid that I may fail at a new assignment or undertaking even though I generally do well at what I attempt.

Not At All True	1	2	3	4	5	Very True

When I've succeeded at something and received recognition for my accomplishments, I have doubts that I can keep repeating that success.

Not At All True	1	2	3	4	5	Very True

If I receive a great deal of praise and recognition for something I've accomplished, I tend to discount the importance of what I have done.

Not At All True	1	2	3	4	5	Very True

I often compare my ability to those around me and think they may be more intelligent than I am.

Not At All True	1	2	3	4	5	Very True

I often worry about not succeeding with a project or on an examination, even though others around me have considerable confidence that I will do well.

Not At All True	1	2	3	4	5	Very True

If I'm going to receive a promotion or gain recognition of some kind, I hesitate to tell others until it is an accomplished fact.

Not At All True	1	2	3	4	5	Very True

I feel bad and discouraged if I'm not "the best" or at least "very special" in situations that involve achievement.

Not At All True	1	2	3	4	5	Very True

Appendix C

Iowa Netherlands Comparison Orientation Measure

Most people compare themselves from time to time with others. For example, they may compare the way they feel, their opinions, their abilities, and/or their situation with those of other people. There is nothing particularly "good" or "bad" about this type of comparison, and some people do it more than others. We would like to find out how often you compare yourself with other people. To do that we would like you to indicate how much you agree with each statement below, by using the scale provided.

I often compare how my loved ones (boy or girlfriend, family members, etc.) are doing with how others are doing.

I disagree strongly.	A	B	C	D	E	**I agree strongly.**

I always pay a lot of attention to how I do things compared with how others do things.

I disagree strongly.	A	B	C	D	E	**I agree strongly.**

If I want to find out how well I have done something, I compare what I have done with how others have done.

I disagree strongly.	A	B	C	D	E	**I agree strongly.**

I often compare how I am doing socially (e.g., social skills, popularity) with other people.

I disagree strongly.	A	B	C	D	E	**I agree strongly.**

I am not the type of person who compares often with others.

I disagree strongly.	A	B	C	D	E	**I agree strongly.**

I often compare myself with others with respect to what I have accomplished in life.

I disagree strongly.	A	B	C	D	E	**I agree strongly.**

I often like to talk with others about mutual opinions and experiences.

I disagree strongly.	A	B	C	D	E	**I agree strongly.**

I often try to find out what others think who face similar problems as I face.

I disagree strongly.	A	B	C	D	E	**I agree strongly.**

I always like to know what others in a similar situation would do.

I disagree strongly.	A	B	C	D	E	**I agree strongly.**

If I want to learn more about something, I try to find out what others think about it.

I disagree strongly.	A	B	C	D	E	**I agree strongly.**

I never consider my situation in life relative to that of other people.

I disagree strongly.	A	B	C	D	E	**I agree strongly.**

Appendix D

Self-Designed Demographics and LinkedIn Usage Questionnaire

How old are you?

What is your gender?

Please Select...

What is your ethnicity?

Please Select...

What is your occupation (if you are still studying put down student)?

How many years of education have you completed?

Do you use LinkedIn?

Please Select...

For how many years have you been using LinkedIn?

On average how many hours per month do you spend on LinkedIn?

On average how many hours per month do you spend browsing your newsfeed on LinkedIn?

On average how many hours per month do you spend finding new connections on LinkedIn?

On average how many hours per month do you spend looking at job postings on LinkedIn?

On average how many hours per month do you spend looking at other people's profiles on LinkedIn?

On average how many hours per month do you spend messaging with other people on LinkedIn?

On average how many hours per month do you spend posting on LinkedIn?

On average how many hours per month do you spend checking likes on your posts on LinkedIn?

On average how many hours per month do you spend updating information on your profile on LinkedIn?

Is there any other activity in which you regularly engage on LinkedIn? If so please specify what it is and the average monthly time you spend on it.

How many connections do you have on LinkedIn?

Would you say the people you follow on LinkedIn are generally more, less or equally successful as you?

Please Select...

Appendix E

Means, Standard Deviation and Correlations for All Measures of LinkedIn Usage and Impostor Score

Table 1

Means, standard deviations, and correlations with confidence intervals

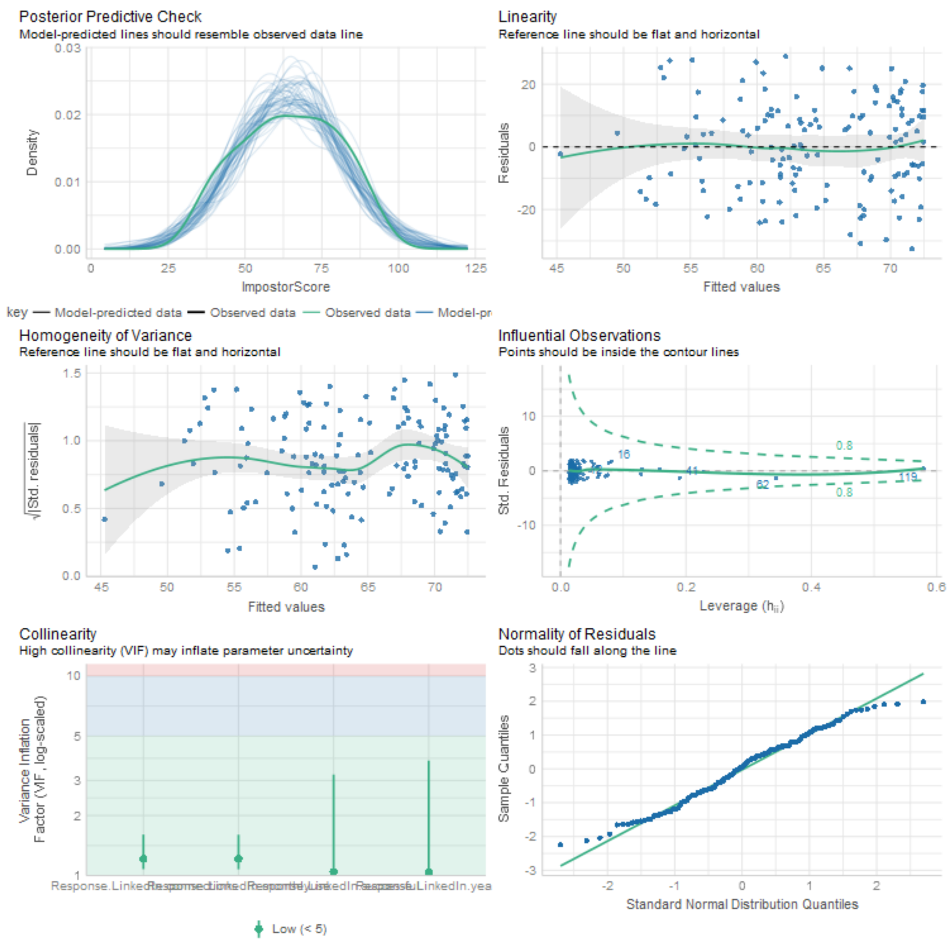
Variable	*M*	*SD*	1	2	3	4	5	6	7	8
1. ImpostorScore	64.20	15.91								
2. Response.LinkedIn.monthlyuse	8.10	14.64	-.04							
			[-.20, .13]							
3. Response.LinkedIn.connections	448.94	705.82	-.13	.40**						
			[-.29, .04]	[.26, .53]						
4. Response.LinkedIn.findingconnections	1.54	3.61	-.01	.43**	.24**					
			[-.17, .16]	[.29, .56]	[.08, .39]					
5. Response.LinkedIn.jobpostings	1.68	3.50	.06	.29**	-.02	.58**				
			[-.10, .22]	[.13, .43]	[-.19, .14]	[.45, .68]				
6. Response.LinkedIn.likes	0.44	2.71	.01	.29**	-.04	.70**	.64**			
			[-.15, .17]	[.13, .43]	[-.20, .12]	[.61, .78]	[.54, .73]			
7. Response.LinkedIn.messaging	0.73	2.69	-.03	.27**	.03	.72**	.66**	.90**		
			[-.19, .14]	[.11, .42]	[-.14, .19]	[.63, .79]	[.55, .74]	[.86, .93]		
8. Response.LinkedIn.newsfeed	3.47	8.76	.01	.57**	.08	.62**	.28**	.43**	.29**	
			[-.15, .18]	[.45, .67]	[-.09, .24]	[.50, .71]	[.12, .42]	[.28, .55]	[.13, .43]	
9. Response.LinkedIn.othersprofiles	2.89	10.80	-.12	.82**	.36**	.18*	.04	.16*	.19*	.16
			[-.28, .05]	[.76, .87]	[.21, .50]	[.01, .33]	[-.13, .20]	[.00, .32]	[.03, .35]	[-.01, .31]
10. Response.LinkedIn.ownprofile	0.77	3.37	-.00	.18*	-.06	.67**	.71**	.94**	.94**	.27**
			[-.17, .16]	[.01, .33]	[-.22, .10]	[.57, .75]	[.61, .78]	[.91, .95]	[.91, .95]	[.11, .42]
11. Response.LinkedIn.posting	0.43	2.58	-.04	.25**	-.00	.73**	.67**	.95**	.95**	.37**
			[-.20, .13]	[.09, .39]	[-.17, .16]	[.65, .80]	[.57, .75]	[.93, .97]	[.93, .97]	[.22, .51]
12. Response.LinkedIn.successful	0.54	0.51	.30**	.14	.07	-.05	.08	-.05	-.04	-.03
			[.14, .44]	[-.03, .30]	[-.10, .23]	[-.21, .11]	[-.09, .24]	[-.21, .11]	[-.21, .12]	[-.19, .14]

13. Response.LinkedIn.years	4.59	4.03	-.26**	.02	.12	.09	-.05	.02	-.01	.24**
			[-.41, -.10]	[-.14, .19]	[-.04, .28]	[-.07, .26]	[-.22, .11]	[-.14, .19]	[-.17, .15]	[.08, .39]

Note. *M* and *SD* are used to represent mean and standard deviation, respectively. Values in square brackets indicate the 95% confidence interval for each correlation. The confidence interval is a plausible range of population correlations that could have caused the sample correlation (Cumming, 2014). * indicates $p < .05$. ** indicates $p < .01$.

Appendix F

Graphs Verifying Linear Regression Model Assumptions on This Study's Data



Appendix G

Results of Moderation Analyses of Each Demographic Variable on the Relationship Between LinkedIn Usage and Impostor Phenomenon

Table 5

Regression results using ImpostorScore as the criterion

Predictor	*b*	*b* 95% CI [LL, UL]	sr^2	sr^2 95% CI [LL, UL]	Fit
(Intercept)	69.80**	[53.64, 85.96]			
Response.LinkedIn.years	0.52	[-1.95, 2.98]	.00	[-.01, .01]	
Response.age	-0.13	[-0.82, 0.56]	.00	[-.01, .01]	
Response.LinkedIn.years:Response.age	-0.03	[-0.10, 0.04]	.00	[-.02, .03]	
					R^2 = .086** 95% CI[.01,.17]

Note. A significant *b*-weight indicates the semi-partial correlation is also significant. *b* represents unstandardized regression weights. sr^2 represents the semi-partial correlation squared. *LL* and *UL* indicate the lower and upper limits of a confidence interval, respectively.
* indicates p < .05. ** indicates p < .01.

Table 6

Regression results using ImpostorScore as the criterion

Predictor	*b*	*b* 95% CI [LL, UL]	sr^2	sr^2 95% CI [LL, UL]	Fit
(Intercept)	60.32**	[46.16, 74.48]			
Response.LinkedIn.years	-1.37	[-3.43, 0.68]	.01	[-.02, .04]	
Response.gender	4.81	[-3.37, 12.99]	.01	[-.02, .04]	
Response.LinkedIn.years:Response.gender	0.32	[-0.96, 1.60]	.00	[-.01, .01]	
					R^2 = .105** 95% CI[.02,.19]

Note. A significant *b*-weight indicates the semi-partial correlation is also significant. *b* represents unstandardized regression weights. sr^2 represents the semi-partial correlation squared. *LL* and *UL* indicate the lower and upper limits of a confidence interval, respectively.
* indicates p < .05. ** indicates p < .01.

Table 7

Regression results using ImpostorScore as the criterion

Predictor	*b*	*b* 95% CI [LL, UL]	sr^2	sr^2 95% CI [LL, UL]	Fit
(Intercept)	75.26**	[63.22, 87.30]			
Response.LinkedIn.years	-3.12	[-6.76, 0.51]	.02	[-.02, .06]	
Response.ethnicity	-1.79	[-5.17, 1.59]	.01	[-.02, .03]	
Response.LinkedIn.years:Response.ethnicity	0.55	[-0.38, 1.49]	.01	[-.02, .04]	
					R^2 = .077* 95% CI[.00,.16]

Note. A significant *b*-weight indicates the semi-partial correlation is also significant. *b* represents unstandardized regression weights. sr^2 represents the semi-partial correlation squared. *LL* and *UL* indicate the lower and upper limits of a confidence interval, respectively.
* indicates p < .05. ** indicates p < .01.

Table 8

Regression results using ImpostorScore as the criterion

Predictor	*b*	*b* 95% CI [LL, UL]	sr^2	sr^2 95% CI [LL, UL]	Fit
(Intercept)	64.35**	[55.62, 73.08]			
Response.LinkedIn.years	-0.66	[-1.62, 0.30]	.01	[-.02, .05]	
Response.occupation	5.07	[-5.04, 15.18]	.01	[-.02, .03]	
Response.LinkedIn.years:Response.occupation	-0.23	[-1.94, 1.49]	.00	[-.01, .01]	
					R^2 = .077* 95% CI[.00,.16]

Note. A significant *b*-weight indicates the semi-partial correlation is also significant. *b* represents unstandardized regression weights. sr^2 represents the semi-partial correlation squared. *LL* and *UL* indicate the lower and upper limits of a confidence interval, respectively.
* indicates p < .05. ** indicates p < .01.

Table 9

Regression results using ImpostorScore as the criterion

Predictor	*b*	*b* 95% CI [LL, UL]	sr^2	sr^2 95% CI [LL, UL]	Fit
(Intercept)	85.43**	[54.15, 116.72]			
Response.LinkedIn.years	-4.82	[-10.30, 0.67]	.02	[-.02, .06]	
Response.education	-1.04	[-3.02, 0.95]	.01	[-.02, .03]	
Response.LinkedIn.years:Response.education	0.23	[-0.10, 0.55]	.01	[-.02, .05]	
					R^2 = .080** 95% CI[.01,.16]

Note. A significant *b*-weight indicates the semi-partial correlation is also significant. *b* represents unstandardized regression weights. sr^2 represents the semi-partial correlation squared. *LL* and *UL* indicate the lower and upper limits of a confidence interval, respectively.
* indicates p < .05. ** indicates p < .01.